

Polynomial Approximation

Data Approximation

Given $\{(x_i, y_i)\}_{i=1}^m$

Find $g(x)$ s.t. $g(x_i) \approx y_i$

★ Interpolation: $g(x_i) = y_i$

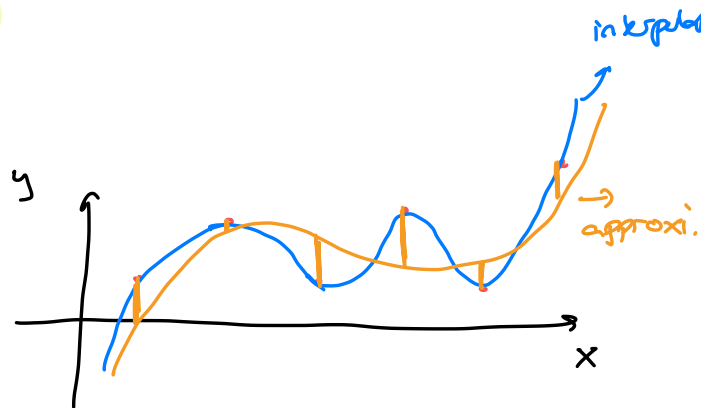
① class of functions

$$g(x) = a_0 \phi_0(x) + a_1 \phi_1(x) + \dots + a_n \phi_n(x)$$

$$g(x) = \underset{\downarrow}{a_0} e^{a_1 x}$$

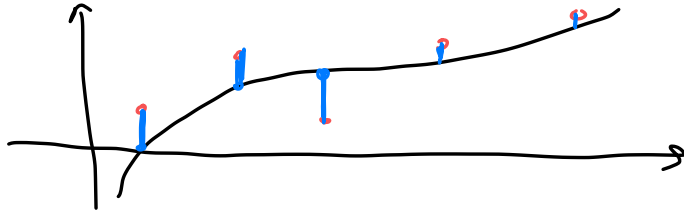
$$\hookrightarrow \phi_j(x) = x^j \leftarrow$$

$$g(x) = a_0 + a_1 x + a_2 x^2 - \dots$$



can be
nonlinear

② Quality of approximation



This is our territory for today

- balance large / small error
- good for not overfitting & noise
- a convex problem

$$- \max_{x_i, i=1, \dots, m} |g(x_i) - y_i|$$

$$- \sum_{i=1}^m (g(x_i) - y_i)^2$$

↳ Least square error (LSE)

$$- \sum_{i=1}^m w_i (g(x_i) - y_i)^2$$

weighted LSE

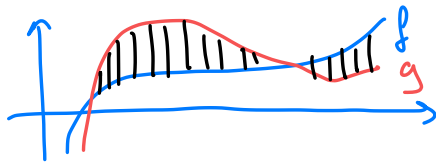
Function Approximation

Given $f: [a, b] \rightarrow \mathbb{R}$

Find g s.t. $g(x) \approx f(x) \quad \forall x \in [a, b]$

$$g(x) = a_0 \phi_0(x) + a_1 \phi_1(x) + \dots + a_n \phi_n(x)$$

Determine the parameters



$$\text{LS Error} = \int_a^b (f(x) - g(x))^2 dx$$

$$\text{wLS error} = \int_a^b w(x) (f(x) - g(x))^2 dx$$

Data Approximation

(Discrete Least Squares)

Given $\{(x_i, y_i)\}_{i=1}^m$

And a parametric f.c. $g(x)$
 $\Rightarrow g(x) = a_0 + a_1 x + \dots$
 $= a_0 \phi_0(x) + \dots$

Determine the params st. LSE is minimized

$$E(a_0, a_1, \dots, a_n) = \sum_{i=1}^m (g(x_i) - y_i)^2$$

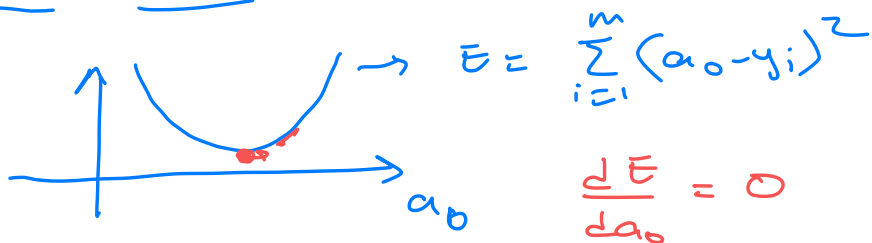
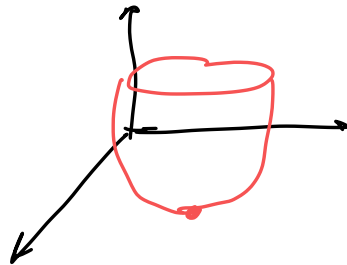
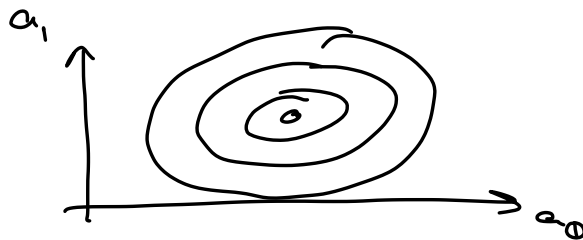
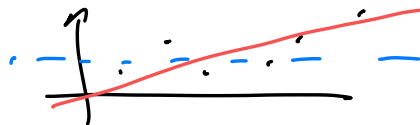
Example: Take $g(x) = a_0 + a_1 x$

$$E(a_0, a_1) = \sum_{i=1}^m (a_0 + a_1 x_i - y_i)^2$$

Optimality condition

$$\frac{\partial E}{\partial a_0} = \sum_{i=1}^m 2(a_0 + a_1 x_i - y_i) = 0$$

$$\frac{\partial E}{\partial a_1} = \sum_{i=1}^m 2(a_0 + a_1 x_i - y_i) x_i = 0$$



$$\frac{dE}{da_0} = 0$$

p

$$\frac{\partial E}{\partial a_0} = \sum_{i=1}^m 2(a_0 + a_1 x_i - y_i) = 0$$

$$\frac{1}{2m} \left(2ma_0 + 2a_1 \sum_{i=1}^m x_i \right) = \left(2 \sum_{i=1}^m y_i \right) \frac{1}{2m}$$

$$a_0 + a_1 \overline{X} = \overline{Y} \quad \leftarrow \quad \overline{X} = \frac{1}{m} \sum_{i=1}^m x_i$$

from

$$\frac{\partial E}{\partial a_1} \Rightarrow \frac{1}{m} \left(a_0 \sum_{i=1}^m x_i + 2a_1 \sum_{i=1}^m x_i^2 \right) = \frac{1}{m} \left(\sum_{i=1}^m x_i y_i \right)$$

$$a_0 \overline{X} + a_1 \overline{X^2} = \overline{XY}$$

$$\hookrightarrow \frac{1}{m} \sum_{i=1}^m x_i^2$$

Two eqns
Two unknowns

General case

$$g(x) = \sum_{j=0}^n a_j \phi_j(x) = a_0 \phi_0(x) + \dots + a_n \phi_n(x)$$

$$E(a_0, \dots, a_n) = \sum_{i=1}^m (a_0 \phi_0(x_i) + a_1 \phi_1(x_i) + \dots + a_n \phi_n(x_i) - y_i)^2$$

Optimality condition

$$\frac{\partial E}{\partial a_j} = \sum_{i=1}^m \cancel{(a_0 \phi_0(x_i) + \dots + a_n \phi_n(x_i) - y_i)} \overbrace{\phi_j(x_i)} = 0$$

$$\sum_{i=1}^m (a_0 \phi_0(x_i) + \dots + a_n \phi_n(x_i)) \phi_j(x_i) = \sum_{i=1}^m y_i \phi_j(x_i)$$

$$\underbrace{\sum_{i=1}^m}_{\substack{\cdot m \\ \text{row}}} a_0 \phi_0(x_i) \phi_j(x_i) + \dots + \underbrace{a_n \sum_{i=1}^m}_{\substack{\cdot m \\ \text{row}}} \phi_n(x_i) \phi_j(x_i) = \quad \downarrow$$

$$\underbrace{\begin{bmatrix} \sum_{i=1}^m \phi_0(x_i) \phi_0(x_i) & \sum_{i=1}^m \phi_0(x_i) \phi_1(x_i) & \dots & \sum_{i=1}^m \phi_0(x_i) \phi_n(x_i) \\ \sum_{i=1}^m \phi_1(x_i) \phi_0(x_i) & \sum_{i=1}^m \phi_1(x_i) \phi_1(x_i) & \dots & \sum_{i=1}^m \phi_1(x_i) \phi_n(x_i) \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^m \phi_n(x_i) \phi_0(x_i) & \sum_{i=1}^m \phi_n(x_i) \phi_1(x_i) & \dots & \sum_{i=1}^m \phi_n(x_i) \phi_n(x_i) \end{bmatrix}}_S \underbrace{\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}}_{\vec{a}} = \underbrace{\begin{bmatrix} \sum_{i=1}^m y_i \phi_0(x_i) \\ \vdots \\ \sum_{i=1}^m y_i \phi_n(x_i) \end{bmatrix}}_{\vec{r}}$$

S

$$S \vec{a} = \vec{r}$$

solve this to get
best coefficients!!!

$$g(x) = a_0^* \phi_0(x) + a_1^* \phi_1(x) + \dots + a_n^* \phi_n(x)$$

$$S_{jk} = \sum_{i=1}^m \phi_j(x_i) \phi_k(x_i)$$

$$r_j = \sum_{i=1}^m y_i \phi_j(x_i)$$

Function Approximation

(continuous least squares)

Approximating function: $g(x) = \sum_{j=0}^n a_j \phi_j(x)$
↪ basis function

Minimize over $a_0, a_1, \dots, a_n$ the weighted LSE

$$E(a_0, \dots, a_n) = \int_a^b w(x) (g(x) - f(x))^2 dx$$

$$= \int_a^b w(x) (a_0 \phi_0(x) + a_1 \phi_1(x) + \dots + a_n \phi_n(x) - f(x))^2 dx$$

optimality condition

$$\frac{\partial E}{\partial a_j} = \int_a^b w(x) (a_0 \phi_0(x) + \dots + a_n \phi_n(x) - f(x)) \phi_j(x) dx = 0$$



Reorganize this

$$a_0 \int_a^b w(x) \phi_0(x) \phi_j(x) dx + \dots + a_n \int_a^b w(x) \phi_n(x) \phi_j(x) dx = \int_a^b w(x) f(x) \phi_j(x) dx$$

↑ we have a eqⁿ like this for
j = 0, ..., n

$$\underbrace{\begin{bmatrix} \int_a^b w(x) \phi_0(x) \phi_0(x) dx & \dots & \int_a^b w(x) \phi_0(x) \phi_n(x) dx \\ \int_a^b w(x) \phi_n(x) \phi_0(x) dx & \dots & \dots \end{bmatrix}}_S \underbrace{\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}}_{\vec{a}} = \underbrace{\begin{bmatrix} \int_a^b w(x) f(x) \phi_0(x) dx \\ \vdots \\ \int_a^b w(x) f(x) \phi_n(x) dx \end{bmatrix}}_{\vec{r}}$$

Solve $S \vec{a} = \vec{r}$ to determine best coefficients.

$$S_{jk} = \int_a^b w(x) \phi_j(x) \phi_k(x) dx$$

$$r_j = \int_a^b w(x) f(x) \phi_j(x) dx$$

- - - - -
If S was diagonal

$$\begin{bmatrix} s_{00} & 0 & \dots & 0 \\ 0 & s_{11} & \dots & 0 \\ & & \ddots & \\ 0 & & & s_{nn} \end{bmatrix} \begin{bmatrix} a_0 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} r_0 \\ \vdots \\ r_n \end{bmatrix}$$

$$a_0 = r_0 / s_{00}$$

$$a_j = r_j / s_{jj}$$

$$\phi_j(x) = x^j$$

$$g(x) = a_0 + a_1 x + a_2 x^2 + \dots$$

— — —

Pick a basis so that S is diagonal

Orthogonal Bases

Basis fnc's $\phi_0, \dots, \phi_n$ are called orthogonal if

$$\rightarrow \int_a^b w(x) \phi_j(x) \phi_k(x) dx = 0 \text{ for } j \neq k$$

over interval $[a, b]$ wrt $w(x)$

Bc. of orthogonality every S_{jk} where $j \neq k$ is 0
so S is diagonal

$$\begin{bmatrix} s_{00} & & \\ & \ddots & \\ & & s_{nn} \end{bmatrix} \begin{bmatrix} a_0 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}$$

$$a_j = \frac{\int_a^b w(x) f(x) \phi_j(x) dx}{\int_a^b w(x) \phi_j(x)^2 dx}$$

↓ doesn't depend on f
so can be precomputed

Generating Orthogonal Bases

$$\phi_0(x) = 1 \quad \leftarrow$$

$$\phi_1(x) = x - B_1$$

$$\phi_2(x) = x \phi_1(x) - B_2 \phi_1(x) - C_2 \phi_0(x)$$

$$\vdots$$
$$\phi_k(x) = (x - B_k) \phi_{k-1}(x) - C_k \phi_{k-2}(x)$$

$$B_k = \frac{\int_a^b x \omega(x) (\phi_{k-1}(x))^2 dx}{\int_a^b \omega(x) (\phi_{k-1}(x))^2 dx} \quad \leftarrow$$

$$C_k = \frac{\int_a^b x \omega(x) \phi_{k-1}(x) \phi_{k-2}(x) dx}{\int_a^b \omega(x) (\phi_{k-2}(x))^2 dx} \quad \leftarrow$$

Legendre Poly. $\rightarrow [-1, 1] \rightarrow \text{interval}$
 $w(x) = 1$

$$P_0(x) = 1$$

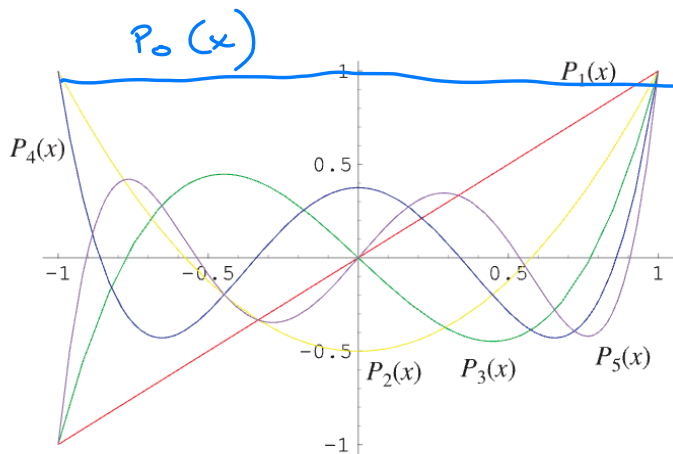
$$P_1(x) = x - B_1 = x \rightarrow B_1 = \frac{\int_{-1}^1 x \overbrace{P_0(x)^2}^1 dx}{\int_{-1}^1 \underbrace{P_0(x)^2}_1 dx} = \frac{\frac{x^2}{2} \Big|_{-1}^1}{x \Big|_{-1}^1} = \frac{0}{2} = 0$$

$$P_2(x) = x^2 - 1/3$$

$$g(x) = \underbrace{a_0}_{\text{circled}} P_0(x) + a_1 P_1(x) + \dots + a_n P_n(x)$$

$$\hookrightarrow a_0 = \frac{\int_{-1}^1 P_0(x) f(x) dx}{\int_{-1}^1 P_0(x)^2 dx}$$

Legendre Polynomials



Normalized Polynomials

$$P_k(1) = 1 \quad \text{for all } k$$

$$\hookrightarrow P_k(x) = \frac{(2k-1)xP_{k-1}(x) - (k-1)P_{k-2}(x)}{k}$$

$$\Rightarrow \int_{-1}^{+1} P_k^2(x) dx = \frac{2}{2k+1} = \frac{1}{k+1/2}$$

$$\underline{\underline{c_k = (k+1/2) \int_{-1}^{+1} P_k(x) f(x) dx}}$$

$$\rightarrow P_0(x) = 1$$

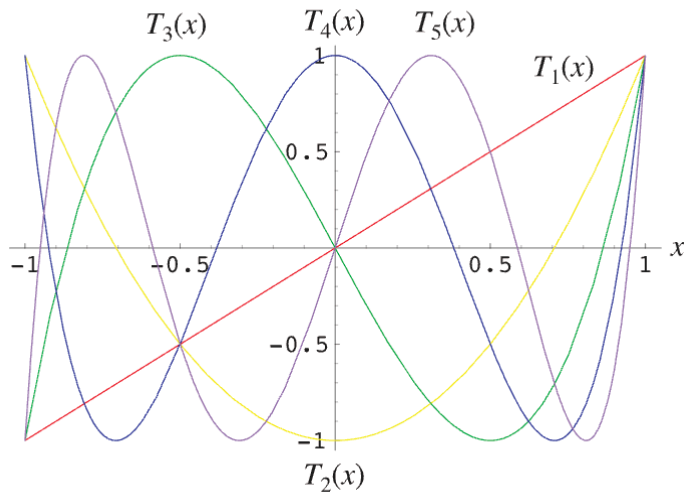
$$\rightarrow P_1(x) = x$$

$$\rightarrow P_2(x) = \frac{3}{2}x^2 - \frac{1}{2}$$

$$\rightarrow P_3(x) = \frac{5}{2}x^3 - \frac{3}{2}x$$

⋮

Chebyshev Polynomials



Interval $[-1, 1]$

$$w(x) = \frac{1}{\sqrt{1-x^2}}$$

$$T_0(x) = 1$$

$$T_1(x) = x$$

$\vdots$

$$\hookrightarrow T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x)$$

Direct formula

$$T_k(x) = \cos(k \arccos(x))$$

$$c_0 = \frac{1}{\pi} \int_{-1}^1 w(x) f(x) dx = \frac{1}{\pi} \int_0^\pi f(\cos \theta) d\theta$$

$$c_k = \frac{2}{\pi} \int_{-1}^1 w(x) T_k(x) f(x) dx = \frac{2}{\pi} \int_0^\pi \cos(k\theta) f(\cos \theta) d\theta$$

Error :

$$\int_a^b w(x) \underbrace{(f(x) - g(x))^2}_{\text{coefficient}} dx = \int_a^b w(x) f(x)^2 dx - \sum_{k=0}^n \alpha_k c_k^2$$

$$\alpha_k = \int_a^b w(x) \phi_k(x)^2 dx$$

Fourier Series

$$w(x) = 1$$

Trigonometric basis $f(x)$

Approx. f over $[-\pi, \pi]$ in LSE by a trigonometric poly.

$$S_n(x) = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos(kx) + b_k \sin(kx))$$

approximately
f.c

Find $a_0, a_1, \dots, a_n, b_1, \dots, b_n$ that minimize

$$\int_{-\pi}^{\pi} (f(x) - S_n(x))^2 dx$$

Form an orthogonal bases

$$\int_{-\pi}^{\pi} \cos(mx) \sin(nx) dx = 0$$

$$\int_{-\pi}^{\pi} \cos(mx) \cos(nx) dx = 0 \quad m \neq n$$

$$\int_{-\pi}^{\pi} \sin(mx) \sin(nx) dx = 0 \quad m \neq n$$

$$a_k = \frac{\int_{-\pi}^{\pi} f(x) \underbrace{\cos(kx)}_{\phi_k(x)} dx}{\int_{-\pi}^{\pi} \cos(kx)^2 dx \quad \pi} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx) dx$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx) dx$$

If f is continuous

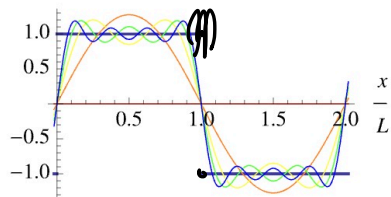
$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx)$$


Error:

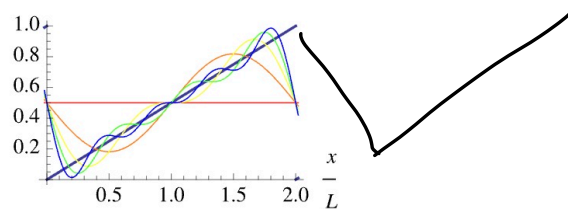
$$\int_{-\pi}^{\pi} (f(x) - s_n(x))^2 dx = \int_{-\pi}^{\pi} f(x)^2 dx - \pi \left(\frac{a_0^2}{2} + \sum_{k=1}^n (a_k^2 + b_k^2) \right)$$

$$= \pi \sum_{k=n+1}^{\infty} (a_k^2 + b_k^2)$$

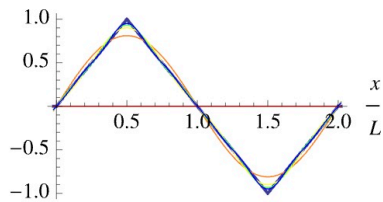
square wave



sawtooth wave



triangle wave



semicircle

